

HARYANA LEET 2026 — FORMULA & QUICK REVISION SHEET

Section C: Mechanical Stream Courses | Questions 51 – 70 | 20 Marks

90 Questions · 90 Minutes · +1 / No Negative Marking · English Medium · Online MCQ

c-1 Mech. Engg. (8Q)

c-2 Production / Mfg. (6Q)

c-3 Auto Engg. (6Q)

[c-1] ELEMENTS OF MECHANICAL ENGINEERING

Applied Mechanics — Statics & Friction

- Friction force: $f = \mu N$ (μ = coefficient of friction)
- Angle of friction: $\tan \phi = \mu$
- Efficiency of screw jack: $\eta = \tan \alpha / \tan(\alpha + \phi)$
- Centroid of triangle: $h/3$ from base; Semi-circle: $4r/3\pi$ from diameter
- Moment = Force $\times$ perpendicular distance

Projectile Motion

- Range: $R = u^2 \sin 2\theta / g$ (max at $\theta = 45^\circ$)
- Max height: $H = u^2 \sin^2 \theta / 2g$
- Time of flight: $T = 2u \sin \theta / g$

Simple Machines

- MA = Load / Effort | VR = Effort distance / Load distance
- Efficiency $\eta = MA/VR \times 100\%$
- Pulley system, wheel & axle, screw jack — common exam machines

Strength of Materials

- Stress $\sigma = F/A$ (Pa or N/m^2)
- Strain $\epsilon = \Delta L/L$ (dimensionless)
- Young's Modulus $E = \sigma/\epsilon$ (slope of stress-strain in elastic region)
- Temperature stress = $E\alpha\Delta T$ (α = linear expansion coeff)
- Poisson's ratio: $\nu = -\text{lateral strain} / \text{axial strain}$ (typical 0.25–0.33)
- Relation: $G = E/[2(1+\nu)]$ | $K = E/[3(1-2\nu)]$
- Resilience = $\sigma^2/(2E)$ (elastic energy per unit volume at yield)
- Principal stresses on inclined planes: Mohr's circle concept
- SFD & BMD — Simply supported beam, UDL: $M_{\max} = wL^2/8$ at midspan
- Cantilever with point load at free end: $M_{\max} = WL$ at fixed end
- Slender columns — Euler's: $P_{cr} = \pi^2 EI/Le^2$ (Le = effective length)

Thermodynamics

- Zeroth Law: if $A=B$ & $B=C$ (thermally) $\rightarrow A=C$ (basis of temperature measurement)
- 1st Law: $Q = \Delta U + W$ (energy conservation)
- 2nd Law: entropy of universe always increases; heat: hot $\rightarrow$ cold only
- Otto cycle (SI engine): $\eta = 1 - 1/r^{\gamma-1}$ (r = compression ratio, $\gamma=1.4$ for air)
- Diesel cycle: $\eta = 1 - [r_c^{\gamma}-1]/[\gamma(r_c-1) \cdot r^{\gamma-1}]$ (r_c = cut-off ratio)
- Dryness fraction $x = m_{\text{steam}}/(m_{\text{water}} + m_{\text{steam}})$
- Enthalpy of wet steam: $h = h_f + x \cdot h_{fg}$

Modes of Heat Transfer

- Conduction (Fourier): $Q/t = kA(T_1 - T_2)/L$
- Convection (Newton): $Q = hA(T_s - T_\infty)$
- Radiation (Stefan-Boltzmann): $Q = \epsilon \sigma AT^4$ ($\sigma = 5.67 \times 10^{-8}$)

Fluid Mechanics

- $\rho = m/V$ | Specific gravity = $\rho_{\text{fluid}}/\rho_{\text{water}}$
- Newton's viscosity law: $\tau = \mu(du/dy)$
- Bernoulli: $P/\rho g + v^2/2g + z = \text{constant}$ (along streamline)
- Continuity: $A_1 v_1 = A_2 v_2$ (incompressible flow)

■ **Reynolds no.** $Re = \rho v D / \mu$ | $Re < 2000$ laminar | $Re > 4000$ turbulent

■ **Darcy-Weisbach:** $h_f = f L v^2 / (2gD)$ (friction head loss)

- Venturimeter: flow meter using Bernoulli principle at converging section
- Pitot tube: measures stagnation pressure $\rightarrow$ velocity $v = \sqrt{2\Delta P / \rho}$

[c-2] PRODUCTION ENGINEERING / MANUFACTURING PROCESSES

Lathe Operations

■ **Cutting speed:** $v = \pi D N / 1000$ (m/min; $D = \text{mm}$, $N = \text{rpm}$)

■ **Machining time:** $t = L / (f \cdot N)$ ($L = \text{length mm}$, $f = \text{feed mm/rev}$)

- Lathe operations: turning, facing, tapering, threading, boring, knurling, parting
- Tool geometry: rake angle (+ve $\rightarrow$ less cutting force), clearance angle

Drilling & Milling

■ **Drilling time:** $t = L / (f \cdot N)$ (L includes approach)

- Types of drills: twist, core, step, counter-bore, counter-sink
- Up-milling (conventional): cutter moves against feed direction
- Down-milling (climb): cutter moves with feed; better surface finish
- Milling cutters: plain, face, end mill, form, gear cutters

Shaper & Planer

- Shaper: tool reciprocates, work fixed; for flat surfaces, slots
- Planer: work reciprocates, tool fixed; for large flat surfaces

Measuring Instruments

■ **Vernier Caliper LC = 1 MSD – 1 VSD** (usually 0.02mm or 0.05mm)

■ **Micrometer LC = Pitch / No. of divisions = 0.01 mm** (standard)

■ **Sine bar:** $\sin \theta = H/L$ ($H = \text{slip gauge height}$, $L = \text{bar length}$, typ 100/200mm)

- Plug gauge: GO/NOGO for checking holes
- Snap gauge: GO/NOGO for shafts | Ring gauge: for external dimensions

Welding, Soldering & Brazing

- Welding: fusion of base metals; permanent joint
- Arc welding: electric arc between electrode and work
- Gas welding: oxy-acetylene flame (neutral: welding, oxidising: brazing, carburising: hard facing)
- Soldering: filler melts $< 450^\circ\text{C}$; Brazing: filler melts $> 450^\circ\text{C}$

Material Science

- Hardness tests: Brinell (BHN, 10mm ball), Rockwell (HRC/HRB), Vickers (VHN/HV, diamond)
- Annealing: heat+slow cool $\rightarrow$ softens, relieves stress, refines grain
- Normalizing: air cool $\rightarrow$ refines grain, higher strength than annealing
- Hardening: heat to austenite+quench $\rightarrow$ martensite (very hard, brittle)
- Tempering: reheat after hardening $\rightarrow$ reduces brittleness
- Carburizing: surface case-hardening by adding carbon to mild steel surface

Engineering Drawing — Key Concepts

- 1st angle (European): object between observer and plane; view on opposite side
- 3rd angle (American): plane between observer and object; view on same side
- Isometric: 30° axes, true lengths shown; hidden lines usually omitted
- Orthographic: 3 views (front, top, side) showing true shape/size

Foundry & Casting

- Pattern types: solid, split, match plate, cope & drag, shell, loose piece
- Pattern allowances: shrinkage, draft (taper), machining, distortion
- Moulding sand: silica (refractoriness) + clay (bond) + moisture + additives

Plastics & Industrial Management

- Thermoplastic: softens on heating (recyclable) — PE, PP, PVC, Nylon
- Thermoset: hardens permanently — Bakelite, Epoxy, Polyester
- Moulding: injection, compression, blow, extrusion, transfer
- Org. structures: Line, Line & Staff, Functional, Committee

■ **Productivity = Output / Input** | **Quality control: SPC, inspection, ISO**

Power Flow

- Engine → Clutch → Gearbox → Propeller Shaft → Differential → Axle → Wheels

Clutch

- Function: connect/disconnect engine from transmission smoothly
- Types: Single plate friction, Multi plate, Centrifugal, Diaphragm, Fluid coupling
- **Torque transmitted:** $T = \mu WR$ (μ =friction, W =axial load, R =effective radius)

Gearbox

- Types: Sliding mesh (old), Constant mesh, Synchromesh (modern)
- **Gear ratio** = N_{input}/N_{output} | **Low gear** → high torque, low speed
- Overdrive: output faster than input; used for fuel efficiency at highway speed

Differential

- Allows inner and outer wheels to rotate at different speeds during cornering
- Types: Bevel gear, Torsen, Locking differential

Brakes

- Drum brake: internal expanding shoes; self-energizing
- Disc brake: caliper presses pads against disc; better heat dissipation
- Hydraulic brakes: Pascal's law — pressure transmitted equally in all directions
- ABS: Anti-lock Braking System — prevents wheel lock-up during hard braking
- **Braking force** $\propto \mu \times W$ (friction coeff $\times$ weight on wheel)

Steering System & Angles

- Types: Rack & Pinion (small cars, precise), Recirculating Ball (heavy vehicles)
- Castor angle: tilts kingpin rearward for self-centering & stability
- Camber angle: wheel tilt from vertical; +ve outward; reduces load on outer bearing
- KPI (King Pin Inclination): reduces steering effort & provides self-centering
- Toe-in: front of wheels closer together; Toe-out: opposite; prevents tyre scrub

Cooling & Lubrication

- Water cooling: radiator + coolant + pump + thermostat; more efficient
- Air cooling: fins on cylinder; simpler, lighter; used in bikes/small engines
- Wet sump lubrication: oil stored in sump; pump circulates
- Dry sump: separate oil tank; used in racing/high-performance engines
- **SAE viscosity:** 5W-30, 10W-40 (W =winter grade); multigrade oils

Tyres, Wheels & EV/Hybrid

- Radial tyre: cords at 90° to tread; better grip, fuel efficiency — modern cars
- Bias-ply: cords at angle; stiffer; older design
- **Tyre marking:** 185/65 R15 → 185=width(mm), 65=aspect ratio, R=radial, 15=rim(inch)
- E-Vehicle: Battery + Motor + Controller + Regenerative braking; zero tail-pipe emission
- Hybrid: ICE + electric motor; self-charging or plug-in (PHEV)
- Major Indian OEMs: Maruti Suzuki, Tata Motors, Mahindra, Hyundai India, Hero, Bajaj